5 Power Round Solutions

1. **a.** There are many examples of sequences of three consecutive positive integers with exactly one cromulent element. One such sequence is 2, 3, 4. Note that 2 and 4 share the common factor of 2, while 3 is prime. More generally, such a sequence will be of the form $2n$, $2n+1$, $2n+2$ for some positive integer n.

b. There are many examples of sequences of three consecutive positive integers with three cromulent elements. One such sequence is 1, 2, 3. More generally, such a sequence will be of the form $2n-1$, $2n$, $2n+1$ for some positive integer n.

2. **a.** There are many examples of sequences of four consecutive positive integers with exactly one cromulent element. One such sequence is 3, 4, 5, 6. Note that 4 and 6 share the common factor of 2, and that 3 and 6 share the common factor of 3, while 5 is prime and thus cromulent.

b. There are many examples of sequences of four consecutive positive integers with two cromulent elements. One such sequence is 2, 3, 4, 5. Note that 2 and 4 share the common factor of 2, while 3 and 5 are prime and thus cromulent.

3. **a.** There are many examples of sequences of five consecutive positive integers with exactly one cromulent element. One such sequence is 2, 3, 4, 5, 6. Note that 2, 4, and 6 share the common factor of 2, and that 3 and 6 share the common factor of 3, while 5 is prime and thus cromulent.

b. There are many examples of sequences of five consecutive positive integers with two cromulent elements. One such sequence is 4, 5, 6, 7, 8. Note that 4, 6, and 8 share the common factor of 2, while 5 and 7 are prime and thus cromulent.

c. There are many examples of sequences of five consecutive positive integers with three cromulent elements. One such sequence is 1, 2, 3, 4, 5. Note that 2 and 4 share the common factor of 2, while 3 and 5 prime, thus 1, 3, and 5 are cromulent.

4. Two consecutive odd integers differ by 2. Thus any common divisor of the two integers must also divide 2. However, the only prime divisor of 2 is 2, and neither of the consecutive odd integers is a multiple of 2. Therefore the consecutive odd integers are relatively prime to each other.

5. **a.** Consider the sequence $a-1$, a, $a+1$. Note that a is relatively prime to $a-1$ and $a+1$ because any common divisor must divide their difference, and the differences are both 1. Thus a is a cromulent element of the sequence.

b. Consider a sequence of four consecutive integers. Notice that two of these integers will be odd, and by problem 4, these are relatively prime to each other. At most one of these two odd integers can be a multiple of 3. Let a be an odd integer in the sequence that is not a multiple of 3. Then a is cromulent by the following reasoning. Note that the difference between a and any other element of the sequence is at most 3. Thus if a shared a common factor greater than 1 with some other element of the sequence, this factor would have to be 2 or 3. But because a is odd and not a multiple of 3, it follows that a is cromulent.

c. Consider a sequence of five consecutive integers. Exactly one number in such a sequence will be a multiple of 5, but that number could also be a multiple of 2 and hence share a common factor with at least one other number in the sequence. There are several cases to consider, namely whether the sequence starts with an even number or an odd number.

If the sequence starts with an even number, then the second and fourth numbers are both odd, and at least one of them is not a multiple of 3 and hence is relatively prime to all other numbers in the sequence because it is neither a multiple of 2 nor 3 and hence is at least 5 away from the nearest integer with a common factor.

If the sequence starts with an odd number, then it again contains an odd number that is not a multiple of 3 and hence is relatively prime to all other numbers in the sequence. In fact, it contains two such numbers if the first or last number is a multiple of 3, and if the middle number is a multiple of 3 then all three odd elements are cromulent.

6. **a.** The minimum number is 1 and the maximum number is 2. One example of a sequence of length 6 with 1 cromulent element is 5, 6, 7, 8, 9, 10, where 7 is the cromulent element. To show that it is not possible for a sequence of 6 consecutive elements to have 0 cromulent elements, consider two cases. If the sequence

ARML encourages the reproduction of our contest problems for non-commercial, educational purposes.
Commercial usage of ARML problems without permission and posting entire contests or contest books are prohibited.

begins with an even number, that number is not cromulent, and one of the other five elements must be cromulent by the argument in the solution to 5(c). A similar argument establishes that one element must be cromulent if the sequence of 6 begins with an odd number (and thus ends in an even number).

One example of a sequence of length 6 with 2 cromulent elements is 1, 2, 3, 4, 5, 6, where 1 and 5 are both cromulent.

To prove that a sequence of length 6 cannot have three cromulent elements, consider that the cromulent elements would all have to be odd, and one of those three would be a multiple of 3. Because one of the even elements must also be a multiple of 3, it is not possible for all three odd elements to be cromulent.

b. The minimum number is 1 and the maximum number is 3. One example of a sequence of length 7 with 1 cromulent element is 4, 5, 6, 7, 8, 9, 10, where 7 is the cromulent element. To show that it is not possible for such a sequence to have zero cromulent elements, consider two cases. If the sequence begins with an even number, then it contains 3 odd numbers. At most one of these is divisible by 3, and at most one is divisible by 5, so one of the odd numbers must be divisible by neither 3 nor 5. This odd number differs by at most 6 from each other element of the sequence, so the only prime factors it can share with another element of the sequence are 2, 3, and 5. Because it is divisible by none of these primes, it follows that the odd number in question is cromulent. Similarly, if the sequence begins with an odd number, then it contains 4 odd numbers; at most two of these are divisible by 3, and at most one is divisible by 5, so again, one odd number in the sequence must be divisible by neither 3 nor 5. By the same argument, this element is cromulent.

One example of a sequence of length 7 with 3 cromulent elements is 1, 2, 3, 4, 5, 6, 7, where 1, 5, and 7 are all cromulent.

To prove that a sequence of length 7 cannot have four cromulent elements, consider that the cromulent elements would all have to be odd. At least one of these four odd elements must be a multiple of 3. Because one of the even elements must also be a multiple of 3, it is thus not possible for all four odd elements to be cromulent.

7. **a.** In any sequence of eight consecutive integers, four of them will be even and hence not cromulent. The primes 3, 5, and 7 are the only numbers that could possibly divide more than one number in the sequence, with 5 and 7 each dividing at most one of the four odd numbers. Note that for 7 to divide two numbers in the sequence, they must be the first and last numbers because only those differ by 7. The prime 3 can divide two odd numbers in the sequence, but only if they differ by 6, which means they must be either the 1st and 7th or the 2nd and 8th numbers in the list. Therefore one of the two odd multiples of 3 would also be the only candidate to be an odd multiple of 7 when there are two multiples of 7, and so at least one of the four odd numbers must be cromulent.

b. In a sequence of nine consecutive integers, again the only primes to check are 2, 3, 5, and 7. If there are five even numbers in the sequence, then at least one of the odd numbers is cromulent by similar reasoning to the solution to 7(a). The difference is that in this case the multiples of 7 can be either the 1st and 8th or the 2nd and 9th numbers, while the odd multiples of 3 must be the 2nd and 8th numbers (because the 1st number is even) and hence overlap with the multiples of 7 again. If instead there are four even numbers in the sequence, then the 1st, 3rd, 5th, 7th, and 9th numbers are odd. Now 7 divides at most one of these odd numbers, 5 divides at most one, and 3 divides at most two, leaving at least one odd cromulent element.

8. **a.** For a sequence of length 17, the primes to check are 2, 3, 5, 7, 11, and 13. By making the first number a multiple of 2, 3, 7, and 13, and the last number a multiple of 2, 5, and 11, such a sequence can be $2 \cdot 3 \cdot 7 \cdot 13 \cdot a$, $5b$, $2c$, $3d$, $2e$, $11f$, $2 \cdot 3 \cdot 5 \cdot g$, $7h$, $2i$, $3j$, $2k$, 5ℓ, $2 \cdot 3 \cdot m$, $13 \cdot n$, $2 \cdot 7 \cdot p$, $3q$, $2 \cdot 5 \cdot 11 \cdot r$ for positive integers $a, b, c, \ldots, r$, so it contains no cromulent elements. Because the 2nd and 6th elements of the above sequence are multiples of 5 and 11, respectively, it follows that the first element of the sequence must be a multiple of 546 that is also 4 (mod 5) and 6 (mod 11), so 2184 works, as does $2184 + 30030s$ for any positive integer s by the Chinese Remainder Theorem. Similarly, the first number could be a multiple of 2, 5, and 11, and the last number a multiple of 2, 3, 7, and 13, which means such a sequence could start with 27830 or $27830 + 30030t$ for any positive integer t.

b. From the argument in the solution to 8(a), the least possible element of such a sequence of length 17 is 2184.

ARML encourages the reproduction of our contest problems for non-commercial, educational purposes.
Commercial usage of ARML problems without permission and posting entire contests or contest books are prohibited.

9. **a.** Let the sequence of k integers be $a_1, a_2, \ldots, a_k$, where $a_n = a_1 + (n-1)$ for $1 \leq n \leq k$. For any $a_j \in \{a_1, a_2, \ldots, a_{2m}\}$, suppose that a_j is divisible by some $p \in \{p_1, p_2, \ldots, p_s\}$. Then $a_{j+p} = a_j + p$, so a_{j+p} is also divisible by p. Hence neither a_j nor a_{j+p} is cromulent because both numbers are divisible by p (also note that $j + p < 4m \leq k$, so a_{j+p} is one of the k integers in the sequence). Thus none of $a_1, a_2, \ldots, a_{2m}$ are cromulent. Similarly, for any $a_{j'} \in \{a_{2m+1}, a_{2m+2}, \ldots, a_k\}$, suppose that $a_{j'}$ is divisible by some $p' \in \{p_1, p_2, \ldots, p_s\}$. Then $a_{j'-p'} = a_{j'} - p'$ is also divisible by p' (also note that $1 < j' - p' \leq k - 2$, so $a_{j'-p'}$ is one of the k integers in the sequence). Thus none of $a_{2m+1}, a_{2m+2}, \ldots, a_k$ are cromulent, completing the proof.

b. Note that $x \equiv 1 \pmod 2$ implies that x is odd. Because $x \equiv 0 \pmod{p_2 p_3 \cdots p_r}$, write $x = q p_2 p_3 \cdots p_r$ for some odd integer q. Then x is divisible by all of the primes $p_2, \ldots, p_r$. The desired result will first be established for numbers of the form $x + 2y$, with $1 \leq y \leq m$. Because x is divisible by each of $p_2, \ldots, p_r$, it suffices to prove that y is divisible by one of $p_2, \ldots, p_r$ unless y is a power of 2. Note that because $p_2, \ldots, p_r$ are all odd, if y is a power of 2 (including $2^0 = 1$), then $2y$ cannot be divisible by any of $p_2, \ldots, p_r$. On the other hand, if y has an odd factor greater than 1, by the Fundamental Theorem of Arithmetic, y must be divisible by at least one of $p_2, \ldots, p_r$ (note that $p_2 = 3$ and $p_r \leq m$). Finally, because the desired result holds for $x + 2y$, with $1 \leq y \leq m$, it also holds for $x - 2y$ because $x - 2y = (x + 2y) - 4y$, thus completing the proof.

c. The definition of the $\pi(\cdot)$ function implies that the number of primes in the set $\{m+1, m+2, \ldots, 2m\}$ is $\pi(2m) - \pi(m)$. Because $m \geq 75$, the lemma can be applied and implies that

$$\pi(2m) - \pi(m) \geq 2\lfloor \log_2(m) \rfloor + 2 = 2t,$$

hence the desired conclusion follows.

d. Note that the moduli of the given congruences:

$$2, \quad p_2 p_3 \cdots p_r, \quad p_{r+u}, \quad p_{r+t+u} \quad (u \in \{1, \ldots, t\})$$

are pairwise relatively prime. The Chinese Remainder Theorem can therefore be applied to the given system, and implies that the given system of congruences has solutions for x, as desired.

e. Let x be a solution to the system from problem 9(d). Then each element of the subsequence of $2m + 2$ elements

$$x - 2m - 1,\ x - 2m + 1,\ \ldots,\ x - 1,\ x + 1,\ \ldots,\ x + 2m + 1 \tag{$*$}$$

is even, hence no element of $(*)$ is cromulent. The remaining elements of the given sequence are

$$x - 2m,\ x - 2m + 2,\ \ldots,\ x - 2,\ x,\ x + 2,\ \ldots,\ x + 2m - 2,\ x + 2m, \tag{$\dagger$}$$

and are all odd. Note that $(\dagger)$ is precisely the sequence considered in problem 9(b). Let $x' = x \pm 2y$ be an element of $(\dagger)$, where $0 \leq y \leq m$. If y is not a power of 2, then the result of problem 9(b) implies that x' is divisible by one of $p_2, \ldots, p_r$. On the other hand, if $y = 2^w$ for some integer $0 \leq w \leq t-1$, then according to the third congruence of problem 9(d), $x+2y$ is divisible by p_{r+w+1}, and according to the fourth congruence of problem 9(d), $x-2y$ is divisible by $p_{r+t+w+1}$. Also note that because $r+w+1 < r+t+w+1 \leq r+2t \leq s$, it follows that p_{r+w+1} and $p_{r+t+w+1}$ are distinct primes among the s primes less than $2m$. Hence the result of problem 9(a) applies, and thus the given sequence has no cromulent elements.

f. Note that $m = \lfloor \frac{k}{4} \rfloor$ implies that $k \leq 4m + 3$. Now consider any sequence $\mathcal{S}$ of k consecutive integers, taken from the sequence of $4m + 3$ consecutive integers

$$x - 2m - 1,\ x - 2m,\ x - 2m + 1,\ \ldots,\ x - 1,\ x,\ x + 1,\ \ldots,\ x + 2m + 1.$$

As proven in problem 9(e), each of these $4m + 3$ integers is divisible by at least one of the s primes less than or equal to $2m$. When a shortened sequence of k consecutive integers is chosen from among the original $4m+3$ integers, it remains true that each of them is divisible by a prime less than or equal to $2m$. Therefore the result of problem 9(a) again implies that the shortened sequence has no cromulent element, as desired.

ARML encourages the reproduction of our contest problems for non-commercial, educational purposes.
Commercial usage of ARML problems without permission and posting entire contests or contest books are prohibited.